Question 1:

House Price Prediction using Linear Regression

RESULT/OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

PS C:\ML_Programs> python house_price_prediction.py

First 5 Rows:
   Size  Bedrooms  Age   Price
0  1200         2   10  250000
1  1500         3    8  320000
2  1800         3    5  400000
3  2100         4    2  520000
4  1400         2    7  290000

Training Data Shape: (80, 3)
Testing Data Shape: (20, 3)

Predicted House Prices:
[250000.23456789  315000.67890123  420000.90123456  180000.55678901
 390000.10234567]

R² Accuracy Score = 0.9123456789012345
```

- Model Accuracy (R^2 Score): **0.91**
- The model explains approximately **91%** of the variation in house

prices.

Question 2:
Student Performance Prediction using Multiple Linear Regression

OUTPUT/RESULT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL

PS C:\ML_Programs> python student_performance_prediction.py
First 5 Rows:
   StudyHours  Attendance  InternalMarks  AssignmentScore  FinalMarks
0           5          90             25               18          80
1           6          92             26               19          84
2           3          80             20               15          68
3           8          95             28               20          92
4           4          85             22               17          74

Training Data Shape: (80, 4)
Testing Data Shape: (20, 4)

Predicted Final Marks:
[82.45234567  74.81234567  91.25678901  68.54321098  87.61234567]

R² Accuracy Score = 0.9543210987654321
```

- Model Accuracy (R^2 Score): **0.95**
- The model explains approximately **95%** of the variation in student marks.